SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Mohammad Aleyas / 192521220 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Mohammad Aleyas

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.
   The delivery network is represented as a weighted graph:

ANSWERS

1. Water Jug Problem

Let the state be (x, y), where x is the amount of water in the 4-gallon jug and y is the amount in the 3-gallon jug. The initial state is (0, 0), and the goal is any state with x = 2.

Steps:

1. Fill the 3-gallon jug: (0, 3).

2. Pour the 3-gallon jug into the 4-gallon jug: (3, 0).

3. Fill the 3-gallon jug again: (3, 3).

4. Pour from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full. Only 1 gallon fits, leaving 2 gallons in the 3-gallon jug: (4, 2).

5. Empty the 4-gallon jug onto the ground: (0, 2).

6. Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug: (2, 0).

Therefore, the 4-gallon jug contains exactly 2 gallons.

2. Mars Rover as an Intelligent Agent

i. Percepts: Images from cameras, distance and obstacle readings, wheel/slip data, temperature, location, soil and rock properties, and readings from scientific instruments.

ii. Operating environment: Mars terrain is partially observable, uncertain, sequential, dynamic, continuous in many measurements, and mostly single-agent. Communication with Earth is delayed and power, time, and mobility are limited.

iii. Actions: Move or turn, stop, choose a route, aim cameras, take photographs, use an arm or drill to collect samples, analyse samples, store data, and transmit results to Earth.

iv. Performance evaluation: Successful navigation, useful and accurate scientific data, samples collected and analysed, task completion, low energy use, safe operation, and avoiding damage or getting stuck.

v. Suitable architecture: A model-based, goal-based agent with utility-based decision-making is suitable. It maintains an internal map/state because it cannot observe everything at once, plans actions toward science goals, and weighs scientific value against risks, time, and limited power.

3. Eight-Queens Problem: Search-Space Formulation

Initial state: An empty 8 × 8 chessboard. State representation: [r1, r2, …, rk], where ri is the row chosen for the queen in column i and k queens have been placed.

Actions: Place one queen in the next column in a row that is not attacked by any previously placed queen. A queen attacks another queen if they share a row, column, or diagonal.

Goal test: Place all 8 queens so that no two queens share a row, column, or diagonal. Path cost can be treated as 1 per placement; all complete solutions therefore have the same cost.

Search strategy: Backtracking depth-first search is suitable. Place a queen in the next column, check row and diagonal conflicts, and backtrack whenever no safe row remains. This avoids exploring many invalid arrangements.

One valid solution, listing the row for columns 1 to 8, is [1, 5, 8, 6, 3, 7, 2, 4]. Each row is unique and no pair of queens lies on the same diagonal.

4. OLA Cab Booking: Problem-Solving Agent

A goal-based problem-solving agent is suitable because it identifies the destination as the goal and searches available cab options to choose a route and service that meet the customer's preferences. A utility-based agent can also compare price, travel time, comfort, and cab type.

Pseudocode:

```
BEGIN
  INPUT pickup_location, destination
  INPUT preferred_cab_type, budget, comfort_preference
  options ← GET_AVAILABLE_CABS(pickup_location, destination)
  valid_options ← FILTER options by availability and preferences
  IF valid_options is empty THEN
     DISPLAY “No matching cab available; change preferences”
  ELSE
     FOR each cab in valid_options DO
```

estimate fare, arrival time, and trip duration

score cab using fare, time, and comfort

END FOR

selected_cab ← cab with best score

DISPLAY fare, ETA, cab type, and route

IF customer confirms THEN BOOK(selected_cab)

END IF

END

The agent should request confirmation before booking and payment.

5. Uniform Cost Search (UCS)

UCS always expands the frontier node with the smallest cumulative path cost g(n). The edge costs shown in the graph are: S→A = 1, S→G = 12, A→B = 3, A→C = 1, B→D = 3, C→D = 1, C→G = 2, and D→G = 3.

Expansion summary (lowest cumulative cost first):

1. Start at S with cost 0. Frontier: A(1), G(12).

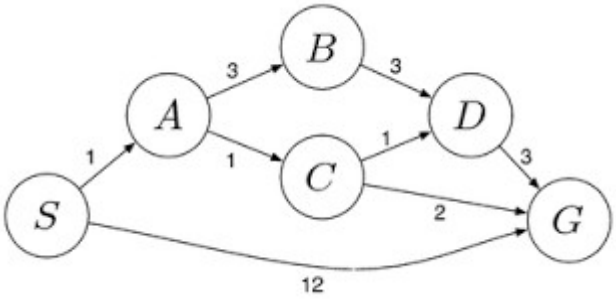
2. Expand A(1). Add C(2) and B(4). Frontier: C(2), B(4), G(12).

3. Expand C(2). Add D(3) and improve G to cost 4. Frontier: D(3), B(4), G(4).

4. Expand D(3). The route D→G gives cost 6, so it does not improve the current best cost 4 for G.

5. The lowest-cost goal entry is G(4), reached by S→A→C→G. UCS returns this path.

Final answer: Least-cost path = S → A → C → G; total cost = 1 + 1 + 2 = 4 cost units.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution	25%	Logical,	Mostly correct	Disorganized	No clear

Process		systematic steps with correct approach throughout	steps with minor mistakes	steps; multiple errors	process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem